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Title: **Simulation assumptions for HAPS co-existence**
Agenda item: 9.13.2.1 Coexistence scenarios and Simulation assumptions
[NR_NTN_solutions-Core]
Document for: Discussion and Approval

1. Introduction

Simulation assumptions for HAPS co-existence study have been captured in [2][1] during RAN4#98bis-e RAN4#99-e respectively.

In this document, previous simulation assumptions [1] are updated to reflect the results of on-going discussions summarized by the moderator [3].

2. Discussion

2.1 Simulation scenarios

The agreed HAPS scenarios for initial coexistence simulations (see the scenario tables in section 1) are (i) rural TN + HAPS, (ii) Urban macro TN + HAPS, and (iii) HAPS + HAPS. All these scenarios use 2 GHz frequency band and FDD duplex scheme.

Considering practical deployment scenarios of HAPS, 20 km altitude for HAPS is assumed when evaluating HAPS + HAPS coexistence in rural environment, since HAPS is intended to serve the rural areas where terrestrial network connectivity is unavailable.

The interference power from the aggressor depends on the distance between the victim and the aggressor. The distance between the victim and aggressor network can be characterized by the center-to-center inter-system distance, which is the distance from the center of the victim network coverage to the center of the aggressor network coverage as shown in Figure 1. Since HAPS antenna gain may vary in the elevation domain the evaluation of coexistence shall be carried out at various center-to-center inter-system distances.

Relevant scenarios for HAPS coexistence study are summarized in Table 1. Note that when the inter-system distance is 0 Km, the HAPS is right above the center of TN network in TN+HAPS coexistence. For HAPS+HAPS coexistence, 0 Km inter-system distance means adjacent channels are operated by the same HAPS.

The simulation scenario is summarized in Figure 1 and Table 1. Note that the TN should consist of 19 sites and 57 sectors as described in Section Error: Reference source not found.

An alternative HAPS and TN layout is proposed in [4], where cluster of TN cells are randomly dropped in the HAPS coverage area as shown in Figure 2. Note that the TN cluster consists of 19 cells, 57 sectors.

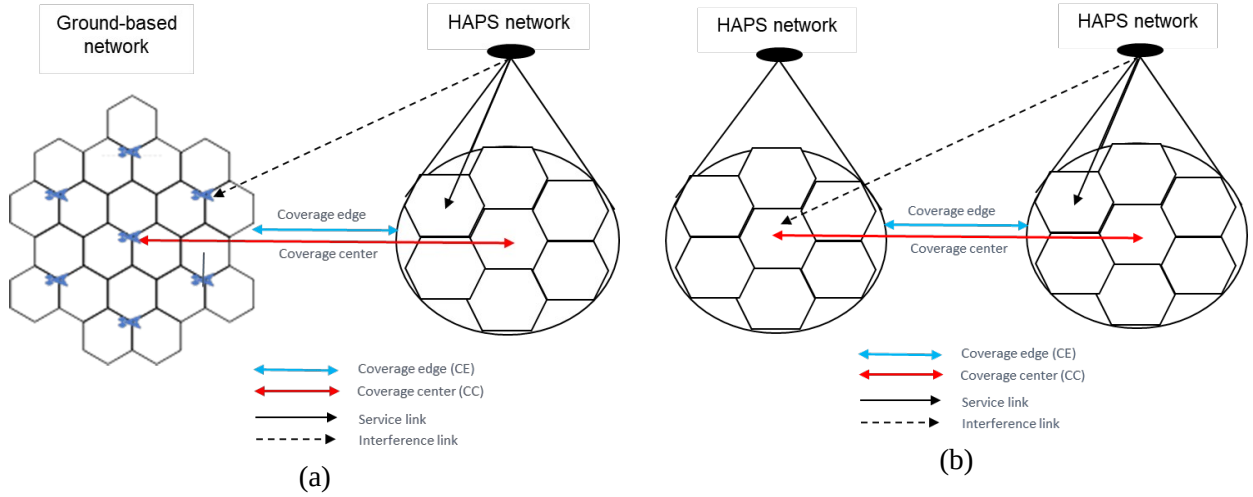


Figure 1. Coexistence scenarios of (a) HAPS and TN, (b) HAPS and HAPS.

Table 1. HAPS coexistence scenarios

HAPS altitude	20 Km
Carrier frequency	2 GHz
Duplex scheme	FDD
Coexistence scenarios	HAPS + TN (UMa)
	HAPS + TN (RMa)
	HAPS + HAPS (RMa)
Center-to-center inter-system distance (Km)	0, 10, 20, 30, 40, 50

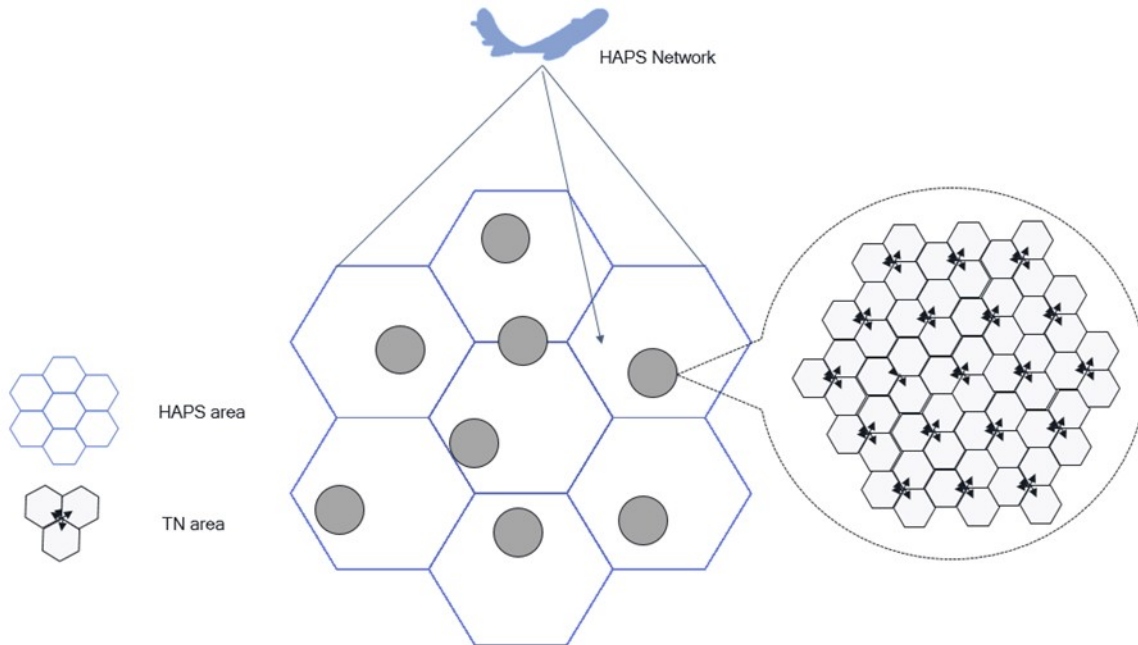


Figure 2. Alternative simulation layout between HAPS and TN

2.2 HAPS antenna and cell layout

A reference HAPS antenna model proposed for HIBS (HAPS as IMT base stations) study in ITU WP-5D [5] is shown in Figure 3. The antenna array is composed of seven antenna panels (six side panels and one

downward facing panel). Antenna elements on each panel are co-phased to form one beam in two crossed linear polarizations to serve one cell. There are a total of seven cells in two layers, one cell in the 1st layer and six cells in the 2nd layer. Other parameters of this antenna model are listed in Table 2. The HAPS antenna model has been revised slightly to adapt to a more realistic implementation, addressing the issues raised in [6].

Although it is possible for HAPS to serve indoor UEs, the large building penetration loss may cause link failure due to low SINR. In a realistic scenario, the majority of the UEs connected to HAPS are outdoor. Therefore, for simplification and as a starting point of the coexistence study all UEs served by HAPS are assumed outdoor UEs.

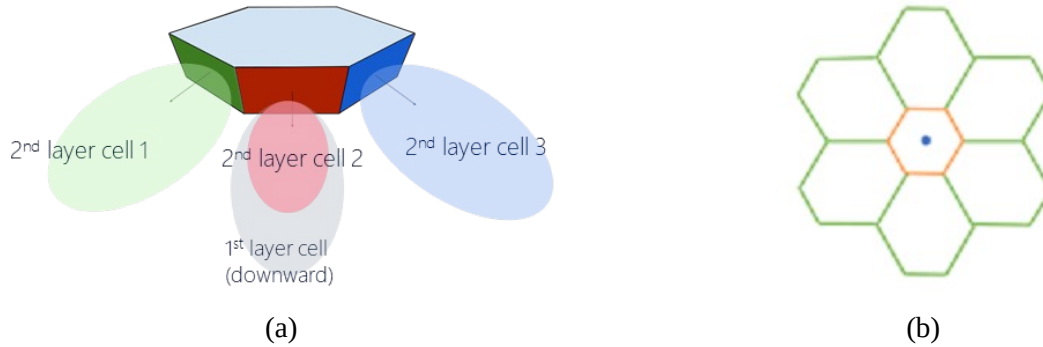


Figure 3. HAPS antenna array and cell layout

Table 2. HAPS network parameters

Number of cells	7
Antenna array configuration (row x column)	2 x 2 for 1st layer cell 4 x 2 for 2nd layer cell
Antenna polarization	Linear $\pm 45^\circ$
Element gain	7.8 dBi
Element HPBW horizontal/vertical	65° for both H/V
Element front-to-back ratio horizontal/vertical	30 dB for both H/V
Element spacing horizontal/vertical	0.7 wavelength for both H/V
Antenna panel tilt (from the horizon)	90° for 1st layer cell 23° for 2nd layer cell
EIPR/cell	56.8 dBm (1st layer cell), 59.8 dBm (2nd layer cell)
EIRP spectral density/cell	43.8 dBm/MHz (1st layer cell), 46.8 dBm/MHz (2nd layer cell)
Tx power per antenna panel	43 dBm
Conducted power (before ohmic loss) per antenna element (dBm)	34 dBm for 2 x 2 (x 2 polarizations) [31] dBm for 4 x 2 (x 2 polarizations)
Noise figure	5 dB
Indoor UE percentage	0%
Coverage area (7 cells combined)	A 100 Km radius circular area centered by the serving HAPS
UE distribution	Uniformly distributed in the coverage area

When HAPS altitude is 20 Km, the antenna gain of this model perceived on the ground is shown in Figure 4, where (a) is the gain of the 1st layer cell (i.e., the center cell) produced by the 90° tilt angle panel (downward facing panel), and (b) is the gain of a 2nd layer cell (i.e., an outer cell) produced by an eastward facing

antenna panel with 23° tilt angle. For a 2nd layer cell, the antenna gain depends on not only the distance but also the azimuth ϕ from the boresight. With this antenna model, SINR can be calculated for a given location on the ground taking into account the propagation loss and co-channel interference. Figure 5 shows the downlink SINR as a function of distance from the coverage center for different azimuth angles ($\phi=0$ is the direction of 2nd layer cell boresight, $\phi=30^\circ$ is at the cell edge), assuming 2 GHz carrier frequency, 20 MHz channel bandwidth, free space path loss, 4 dB fade margin, and 7 dB UE noise figure. It can be observed that for outdoor UEs in a rural environment, where the propagation condition is close to free space path loss with shadow fading, HAPS coverage range can reach 100 km at 2 GHz frequency.

Based on the above a coverage range of 100 km is assumed when operating in the 2 GHz range. UEs should be dropped uniformly in a 100 Km radius circular coverage area centered by the serving HAPS.

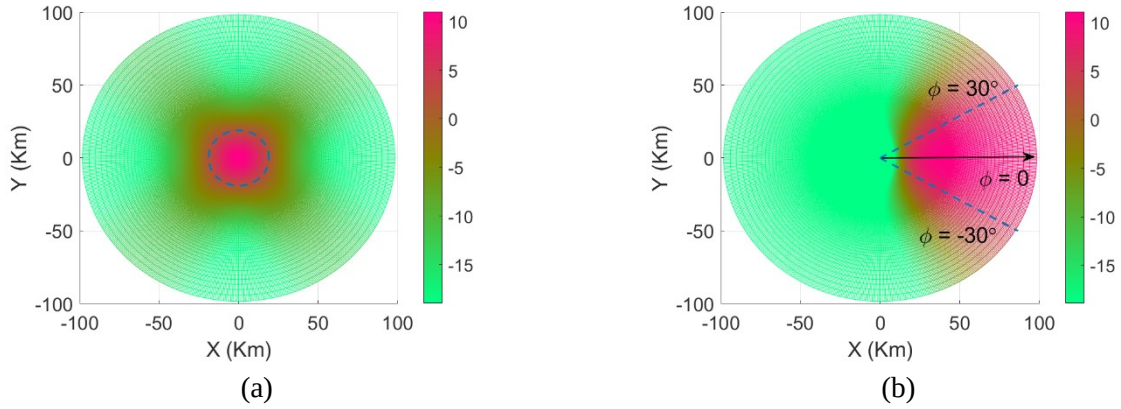


Figure 4. HAPS antenna gain (in dB) in a 100 Km radius area on the ground. (a) Antenna gain of the 1st layer cell. (b) Antenna gain of a 2nd layer cell.

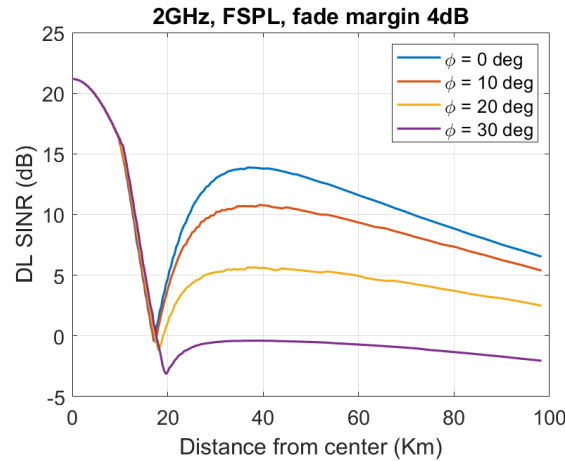


Figure 5. HAPS single system SINR as a function of distance from coverage center assuming 2 GHz carrier frequency, free space path loss and 4 dB fade margin

2.3 Terrestrial network layout

The typical network layout of 19 sites, 3 sectors per site, with co-channel interference wrap-around can be adopted for the coexistence study. System parameters such as inter-site distance, BS antenna height, antenna array and array downtilt angle, indoor UE percentage, etc. should be adjusted according to the environment. Tables 3-5 summarize the assumptions for Urban macro and rural macro environments.

Table 3. Terrestrial network parameters

Terrestrial environment	Urban macro	Rural macro
Network layout	19 sites (57 cells)	19 sites (57 cells)

	wrap-around	wrap-around
Inter-site distance	750 m	7.5 Km
BS antenna height	25 m	30 m
BS transmit power	46 dBm	46 dBm
Conducted power per antenna element	25 dBm	25 dBm
BS antenna array (M, N, P)	(8, 8, 2)	(8, 8, 2)
BS antenna Element spacing horizontal/vertical	H 0.5λ / V 0.7λ	H 0.5λ / V 0.9λ
BS antenna downtilt	10°	3°
BS antenna element gain pattern	Tables 4, 5	Tables 4, 5
BS noise figure	5 dB	5 dB
Indoor UE percentage	[0, 80%]	[0, 50%]
Minimum BS-UE distance	35 m	35 m

Table 4. Terrestrial BS antenna element gain pattern (TR 38.921 [11])

Description	Equation	Unit
Peak normalized element radiation pattern	$A(\theta, \varphi) = -\min \left[-\min \left[12 \left(\frac{\varphi}{\varphi_{3dB}} \right)^2, A_m \right] - \min \left[12 \left(\frac{\theta - 90}{\theta_{3dB}} \right)^2, SLA_v \right], A_m \right]$	dB
Peak gain normalized element radiation pattern	$A_E(\theta, \varphi) = G_{E,max} + A(\theta, \varphi)$	dBi
Composite array radiation pattern	$A_A(\theta, \varphi) = A_E(\theta, \varphi) + 10 \log_{10} \left(\left \sum_{m=1}^M \sum_{n=1}^N w_{m,n} v_{m,n} \right ^2 \right)$, where $v_{m,n} = \exp \left(j 2 \pi \left((m-1) \frac{d_v}{\lambda} \cos(\theta) + (n-1) \frac{d_h}{\lambda} \sin(\theta) \sin(\varphi) \right) \right)$ $w_{m,n} = \frac{1}{\sqrt{MN}} \exp \left(j 2 \pi \left((m-1) \frac{d_v}{\lambda} \sin(\theta_{etilt}) - (n-1) \frac{d_h}{\lambda} \cos(\theta_{etilt}) \sin(\varphi_{escan}) \right) \right)$	dBi

Table 5. Terrestrial BS array antenna parameters

Parameter	Urban Macro	Rural Macro
Element gain ¹ $G_{E,max}$ (dBi)	6.4	7.1
Horizontal front-to-back ratio A_m (dB)	30	30
Vertical front-to-back ratio SLA_v (dB)	30	30
Horizontal 3dB beamwidth φ_{3dB} (deg.)	90	90
Vertical 3dB beamwidth θ_{3dB} (deg)	65	54
Horizontal element spacing $d_h(\lambda)$	0.5	0.5
Vertical element spacing $d_v(\lambda)$	0.7	0.9
Note 1: The element gain includes the Ohmic loss of 2 dB.		

2.4 Propagation model

Consider coexistence scenarios TN+HAPS and HAPS+HAPS. The radio link between a TN BS and an UE served by TN (TN UE) may follow either the UMa model or the RMa model in [9] depending on the terrestrial environment. The radio link between HAPS and a UE, regardless of the UE being served by TN or HAPS, follows the NTN path loss model of either “Urban” or “Rural” scenario. For the HAPS to TN UE link, an additional O-to-I penetration loss needs to be applied if the UE is indoor. This penetration loss is not needed for HAPS UE (i.e., UE served by HAPS) since all HAPS UEs are assumed to be outdoor. The O-to-I penetration loss model specified in [7] with 50%/50% probability for the low-loss/high-loss model is used when applicable. The use of channel model for different radio links in the coexistence scenarios is summarized in Table 6.

Table 6. Channel model used in HAPS coexistence study

Scenario	Radio Link	Channel model	Reference
TN+HAPS (UMa)	TN BS to TN UE	Urban Macro	TR 38.901 [9]
TN+HAPS (UMa)	HAPS to TN UE	NTN Urban + penetration loss ¹	TR 38.811 [8]
TN+HAPS (UMa)	HAPS to HAPS UE	NTN Urban	TR 38.811 [8]
TN+HAPS (RMa)	TN BS to TN UE	Rural Macro	TR 38.901 [9]
TN+HAPS (RMa)	HAPS to TN UE	NTN Rural+ penetration loss ¹	TR 38.811 [8]
TN+HAPS (RMa)	HAPS to HAPS UE	NTN Rural	TR 38.811 [8]
HAPS+HAPS (RMa)	HAPS to HAPS UE	NTN Rural	TR 38.811 [8]
Note 1: Penetration loss model is specified in TR 38.803 [7], assuming 50% low-loss model and 50% high-loss model. It only applies to indoor UEs.			

2.5 UE assumption

The same UE characteristics should be used for both TN UEs and HAPS UEs. It is reasonable to assume that UE has a single omni-directional antenna element with linear cross-polarizations. The UE assumptions is summarized in Table 7.

Table 7. UE assumption

UE antenna array (M, N, P)	(1, 1, 2)
UE antenna element gain	0 dBi, omni-directional
UE transmit power	23 dBm
UE noise figure	9 dB

2.6 Channel bandwidth and scheduled bandwidth

For FDD at 2 GHz band, 20 MHz channel bandwidth and 15 KHz subcarrier spacing is assumed. For downlink simulations, UEs are scheduled in a round-robin fashion and the scheduled UE is given the full bandwidth. For uplink simulations, attention shall be paid to the power limited nature of HAPS networks. When the UE is in a NLOS condition with HAPS, the allocated bandwidth may need to be reduced to the minimum in order to overcome the additional clutter loss (17-19 dB) and maintain an acceptable SINR. For terrestrial NR network, 3 UEs are scheduled in the UL in a round-robin fashion.

A simple and realistic model of UL bandwidth allocation for the considered HAPS coexistence scenarios is still for further discussion. For initial simulations, we can consider parameters in Table 8.

Table 8. Proposed DL and UL transmission bandwidth

Parameters	Downlink	Uplink
Subcarrier spacing (SCS)	15 KHz	15 KHz
Channel bandwidth	20 MHz	20 MHz

Scheduled bandwidth per TN UE	20 MHz	33 RBs
Number of scheduled UEs per TN cell	1	3
Scheduled bandwidth per HAPS UE	20 MHz	TBD
Number of scheduled UEs per HAPS cell	1	[10] or [3]

2.7 Uplink transmission power control model

Since terrestrial NR and HAPS networks have very different coverage and topology, UL power control setting for TN and HAPS should not be the same. The same power control model may be used but the UE transmit power should depend on the allocated bandwidth. To start the initial simulations, we can use the power control model with UE transmit power P_t determined according to

$$P_t = P_{\max} \times \min \left(1, \max \left[R_{\min}, \left(\frac{CL}{CL_{x-ile}} \right)^\gamma \right] \right)$$

where, $P_{\max} = 23\text{dBm}$, $R_{\min} = \text{TBD dB}$, CL_{x-ile} and γ are set as following:

- $CL_{x-ile} = 88 + 10 \cdot \log_{10}(200/X) + 11 - Y$,
where X is UL transmission BW (MHz) and Y is the BS noise figure
- $\gamma = 1$

UEs connected to TN and HAPS networks may have different X (transmission BW) in this model. For TN, 3 UEs are scheduled and each is allocated 33 RBs. For HAPS, UL scheduled bandwidth is still TBD, but 2 RBs can be assumed for calibration purpose. This assumption will be further revised to reflect the latest discussion agreements.

Table 9. UL power control parameters

UL power control parameter	TN	HAPS
$P_{\max}$ (dBm)	23	23
$R_{\min}$ (dB)	-54	-54
γ	1	1
X, transmission bandwidth (MHz)	[5.94]	[0.36]
Y, BS noise figure (dB)	5	5

3. Conclusion

It is proposed to use the simulation assumptions in this paper for NTN HAPS co-existence study.

References

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- [4] R4-2110800, Qualcomm, “HAPS coexistence simulation assumptions,” 3GPP RAN4#99-e.
- [5] SoftBank, Loon LLC, Nokia, Ericsson, “Proposed deployment and system characteristics of HIBS in the working document towards a preliminary draft new Report ITU-R M.[HIBS-CHARACTERISTICS],” ITU WP-5D contribution, Sep. 28, 2020.
- [6] R4-2110119, Ericsson, “NTN: coexistence studies – simulation assumptions discussion,” 3GPP RAN4#98bis-e.
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- [10] 3GPP TR 36.942, “Radio Frequency (RF) system scenarios.”
- [11] 3GPP TR 38.921, “Study on International Mobile Telecommunications (IMT) parameters for 6.425-7.025GHz, 7.025-7.125GHz and 10.0-10.5 GHz.”